VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Reg. Number:	

Continuous Assessment Test (CAT) – II - MARCH 2026

Programme	:	B. Tech (CSE)	Semester	:	Fall 25-26
Course Code & Course Title	:	BCSE301L & Software Engineering	Class Number	:	CH2025260502029 CH2025260502034 CH2025260502039 CH2025260502042
Faculty	:	Dr. AMUTHA S Dr. ALOK CHAUHAN Dr. PRABHAKARAN R Dr. V. SAKTHIVEL	Slot	:	D2+TD2
Duration	:	90 Minutes	Max. Mark		50

General Instructions:

- Write only your registration number on the question paper in the box provided, and do not write other information
- Only a non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1		A university library plans to develop a Library Management System (LMS) to automate its manual operations. Currently, book issuing, returning, fine calculation, and record maintenance are handled manually, leading to errors and delays. The proposed system should allow students and faculty members to search for books, reserve books online, and check their borrowing history. Librarians should be able to add, update, and delete book records. The system must automatically calculate fines for late returns. Only authorized staff members can modify or delete records. The system should maintain transaction logs for auditing purposes. It must support at least 5,000 users and handle 500 concurrent users during peak hours. The system should be available 24/7 except during scheduled maintenance. All user data must be securely stored and protected from unauthorized access	10	CO3	K3
	(i)	Prepare a structured Software Requirements Specification (SRS) document outline for the above system. (5 Marks)			
	(ii)	Identify system constraints, assumptions, and dependencies. (2 Marks)			
	(iii)	Develop a simple use-case description for the 'Book Issue' process. (3 Marks)			

2		You are validating the requirements for a new high-speed ATM. One requirement states: The ATM shall dispense any requested amount of cash instantly and must be compatible with all magnetic and chip-based cards ever manufactured. During a Requirements Validation meeting, a Hardware Engineer notes that the mechanical dispenser requires 3 seconds to move cash to the slot, and a Security Expert points out that "all cards ever manufactured" includes obsolete, insecure cards that the bank's policy strictly prohibits.	10	CO3	K3
	(i)	Identify and explain three specific validation criteria that the "Auto-Cash" requirement fails to satisfy. (5 marks)			
	(ii)	Explain how User Interface (UI) Prototype of the ATM screen could validate a customer's expectation of "transaction speed" better than a written document. (5 marks)			
3		A healthcare management system exposes patient records across all modules. The billing module can view complete patient history and the pharmacy module can modify admission details. This creates risks in security, privacy and maintainability.	10	CO3	K3
	(i)	Using appropriate design concepts such as information hiding, abstraction and modularity, identify the design flaws present in this scenario. (4 marks)			
	(ii)	Redesign the healthcare management system using design concepts such as separation of concerns, refinement, architecture partitioning, and system decomposition (6 marks)			
4		You are designing a system that ingests large volumes of application logs from multiple servers (real-time streaming or batch files), performs a sequence of transformations and analyses (parsing, error filtering, adding metadata, count aggregation, detect anomalies), and stores the results for querying/reporting by dashboards, BI tools, alerting systems, and ad-hoc queries. Design a suitable architecture for each of the following tasks with the help of a neat diagram and explanation:	10	CO3	K4
	(i)	log processing transformation (5 marks)			
	(ii)	interactive querying and multiple tools accessing historical data (5 marks)			

5	(i) (ii) (iii)	A global software organization recently faced customer complaints regarding inconsistent software quality across its products. During an internal quality review, the management realized that development teams were following different documentation practices, coding standards, and review mechanisms. As part of the quality improvement initiative, the organization decided to introduce systematic process evaluation techniques to detect deviations and ensure adherence to development standards. Examine how formal examination methods and independent evaluation mechanisms contribute to improving software process quality. (3 Marks) Differentiate between these two techniques in terms of purpose, procedure, stakeholders involved, and expected results. (2 Marks) Critically assess how the adoption of these practices can help organizations identify process gaps, enforce compliance with development standards, and enhance overall software quality. (5 Marks)	10	CO4	K5

**********All the best **********

Faculty Signature